

61. If in a G.P. of 64 terms, the sum of all the terms is 7 times sum of the odd terms the G.P., then the common ratio of the G.P. is equal to :

यदि 64 पदों की एक G.P. में सभी पदों का योग, इसके विषम पदों के योग का 7 गुना है, तो G.P. का सार्व अनुपात बराबर है:

- (A) 7 (B) 6 (C) 4 (D) 5

Ans. B

Sol. $a + ar + ar^2 + ar^3 + \dots + ar^{63}$

$$= 7(a + ar^2 + ar^4 + \dots + ar^{62})$$

$$\Rightarrow \frac{a(1-r^{64})}{1-r} = \frac{7a(1-r^{64})}{1-r^2}$$

$$r = 6$$

62. Let $A = \begin{bmatrix} 2 & a & 0 \\ 1 & 3 & 1 \\ 0 & 5 & b \end{bmatrix}$. If $A^3 = 4A^2 - A - 21I$, where I is the identity matrix of order 3×3 , then $2a + 3b$ is equal to

:

माना $A = \begin{bmatrix} 2 & a & 0 \\ 1 & 3 & 1 \\ 0 & 5 & b \end{bmatrix}$ यदि $A^3 = 4A^2 - A - 21I$, जहाँ I, 3×3 का तत्समक आव्यूह है, तो $2a + 3b$ बराबर होगा :

- (A) -9 (B) -13 (C) -12 (D) -10

Ans. B

Sol. $\begin{vmatrix} 2-\lambda & a & 0 \\ 1 & 3-\lambda & 1 \\ 0 & 5 & b-\lambda \end{vmatrix} = 0$

$$\Rightarrow (2-\lambda)(3b-3\lambda-\lambda b+\lambda^2-5)-ab+a\lambda=0$$

$$\Rightarrow 6b-6\lambda-2\lambda b+2\lambda^2-10-3\lambda b+3\lambda^2+\lambda^2 b-\lambda^3+5\lambda-ab+a\lambda=0$$

$$\Rightarrow -\lambda^3+\lambda^2(2+2+b)+\lambda(-6-2b-3b+5+a)+6b-10-ab=0$$

$$\Rightarrow -\lambda^3+\lambda^2(b+5)+\lambda(a-5b-1)+6b-10-ab=0$$

$$\Rightarrow -A^3+4A^2-A-21I=0$$

$$b+5=4 \quad a-5b-1=-1$$

$$b=-1$$

$$a=-5$$

$$2a+3b$$

$$\Rightarrow 2(-5)+3(-1)$$

$$\Rightarrow -13$$

63. Let $R = \begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix}$ be a non-zero 3×3 matrix, where

$x \sin \theta = y \sin \left(\theta + \frac{2\pi}{3} \right) = z \sin \left(\theta + \frac{4\pi}{3} \right) \neq 0, \theta \in (0, 2\pi)$. For a square matrix M , let $\text{trace}(M)$ denote the sum of all the diagonal entries of M . Then, among the statements :

(I) $\text{Trace}(R) = 0$

(II) If $\text{trace}(\text{adj}(\text{adj}(R))) = 0$, then R has exactly one non-zero entry.

(A) Both (I) and (II) are true

(B) Only (II) is true

(C) Only (I) is true

(D) Neither (I) nor (II) is true

माना $R = \begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix}$, 3×3 का एक शून्येत्तर आव्यूह है, जहाँ

$x \sin \theta = y \sin \left(\theta + \frac{2\pi}{3} \right) = z \sin \left(\theta + \frac{4\pi}{3} \right) \neq 0, \theta \in (0, 2\pi)$ एक वर्ग आव्यूह M के लिए माना $\text{trace}(M)$, M के विकर्ण के सभी अवयवों के योग को दर्शाता है। तो कथनों :

(I) $\text{Trace}(R) = 0$

(II) यदि $\text{trace}(\text{adj}(\text{adj}(R))) = 0$ है, तो R का केवल एक अवयव शून्येत्तर है

(A) (I) तथा (II) दोनों सत्य हैं

(B) केवल (II) सत्य हैं

(C) केवल (I) सत्य हैं

(D) न तो (I) नहीं (II) सत्य है

Ans. D

Sol. $x \sin \theta = y \sin \left(\theta + \frac{2\pi}{3} \right) = z \sin \left(\theta + \frac{4\pi}{3} \right) \neq 0$

$\Rightarrow x, y, z \neq 0$

Also,

$$\sin \theta + \sin \left(\theta + \frac{2\pi}{3} \right) + \sin \left(\theta + \frac{4\pi}{3} \right) = 0 \forall \theta \in \mathbb{R}$$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$$

$$\Rightarrow xy + yz + zx = 0$$

$$(i) \text{ Trace } (R) = x + y + z$$

$$\text{If } x + y + z = 0 \text{ and } xy + yz + zx = 0$$

$$\Rightarrow x = y = z = 0$$

Statement (i) is false

$$(ii) \text{ Adj } (\text{Adj}(R)) = |R| R$$

$$\text{Trace } (\text{Adj}(\text{Adj}(R)))$$

$$= xyz(x + y + z) \neq 0$$

Statement (ii) is also false.

64. If A , B , and $(\text{adj}(A^{-1}) + \text{adj}(B^{-1}))$ are non-singular matrices of same order, then the inverse of $A(\text{adj}(A^{-1}) + \text{adj}(B^{-1}))^{-1}B$, is equal to :

यदि A , B , और $(\text{adj}(A^{-1}) + \text{adj}(B^{-1}))$ समान कोटि के व्युत्क्रमणीय आव्यूह हैं, तो $A(\text{adj}(A^{-1}) + \text{adj}(B^{-1}))^{-1}B$ का व्युत्क्रम बराबर है :

$$(A) \text{ adj}(B^{-1}) + \text{adj}(A^{-1})$$

$$(B) \frac{AB^{-1}}{|A|} + \frac{BA^{-1}}{|B|}$$

$$(C) AB^{-1} + A^{-1}B$$

$$(D) \frac{1}{|AB|} (\text{adj}(B) + \text{adj}(A))$$

Ans. D

Sol. $(A(\text{adj}(A^{-1}) + \text{adj}(B^{-1}))^{-1}B)^{-1}$

$$= B^{-1} \left(\text{adj} \left(\frac{\text{adj}(A)}{|A|} \right) + \text{adj} \left(\frac{\text{adj}(B)}{|B|} \right) \right) A^{-1}$$

$$= B^{-1} \left(\frac{1}{|A|^{n-1}} |A|^{n-2} A + \frac{1}{|B|^{n-1}} |B|^{n-2} B \right) A^{-1}$$

$$= \frac{\text{adj}(B)}{|B|} \left(\frac{A}{|A|} + \frac{B}{|B|} \right) \frac{\text{adj}(A)}{|A|}$$

$$= \frac{\text{adj}(B) \cdot A \cdot \text{adj}(A)}{|B||A|^2} + \frac{\text{adj}(B) \cdot B \cdot \text{adj}(A)}{|A||B|^2}$$

$$= \frac{\text{adj}(B) \cdot |A| \cdot I}{|B| |A|^2} + \frac{|B| \cdot I \cdot \text{adj}(A)}{|A| |B|^2}$$

$$= \frac{1}{|A| |B|} (\text{adj}(A) + \text{adj}(B))$$

65. Let 3, a, b, c be in A.P. and 3, a - 1, b + 1, c + 9 be in G.P. Then, the arithmetic mean of a, b and c is :

माना 3, a, b, c एक A.P. में हैं तथा 3, a - 1, b + 1, c + 9 एक G.P. में हैं। तो a, b तथा c का समांतर माध्य है :

- (A) -4 (B) 11 (C) 13 (D) -1

Ans. B

Sol. 3, a, b, c be in A.P.

$$a - 3 = b - a = c - b$$

3, a - 1, b + 1, c + 9 in G.P.

$$\frac{a-1}{3} = \frac{b+1}{a-1} = \frac{c+9}{b+1} \quad (a \neq 1, b \neq -1)$$

$$(a-1)^2 = 3(b+1)$$

$$a^2 + 1 - 2a = 3b + 3$$

$$a^2 - 8a + 7 = 0$$

$$(a-1)(a-7) = 0$$

$$a = 7 \text{ as } a \neq 1$$

$$b = 2a - 3$$

$$= 14 - 3$$

$$= 11$$

$$c = 2b - a = 22 - 7 = 15$$

Mean of a, b, c

$$\frac{a+b+c}{3} = \frac{7+11+15}{3} = \frac{33}{3} = 11$$

66. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{bmatrix}$ and $|2A|^3 = 2^{21}$ where $\alpha, \beta \in \mathbb{Z}$. Then a value of α is :

माना $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{bmatrix}$ तथा $|2A|^3 = 2^{21}$ हैं, जहाँ $\alpha, \beta \in \mathbb{Z}$ हैं। तो α का एक मान है :

(A) 5

(B) 17

(C) 3

(D) 9

Ans. A

Sol. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{bmatrix}$

$$|A| = \alpha^2 - \beta^2$$

$$|2A|^3 = 2^{21}$$

$$\Rightarrow 2^9 |A|^3 = 2^{21}$$

$$\Rightarrow |A|^3 = 2^{12}$$

$$|A| = 2^4 \quad \Rightarrow \quad \alpha^2 - \beta^2 = 16$$

$$(\alpha, \beta) = (5, 3)$$

$$\alpha = 5$$

67. Let the system of equations :

$$2x + 3y + 5z = 9,$$

$$7x + 3y - 2z = 8,$$

$$12x + 3y - (4 + \lambda)z = 16 - \mu,$$

have infinitely many solutions. Then the radius of the circle centred at (λ, μ) and touching the line $4x = 3y$ is :

माना समीकरण निकाय

$$2x + 3y + 5z = 9,$$

$$7x + 3y - 2z = 8,$$

$$12x + 3y - (4 + \lambda)z = 16 - \mu,$$

के अनन्त हल हैं। तो उस वृत्त, जिसका केन्द्र (λ, μ) है और जो रेखा $4x = 3y$ को स्पर्श करता है, की त्रिज्या है :

(A) 7

(B) $\frac{17}{5}$

(C) $\frac{21}{5}$

(D) $\frac{7}{5}$

Ans. D

Sol. $D = \begin{vmatrix} 2 & 3 & 5 \\ 4 & 3 & -2 \\ 12 & 3 & -(\lambda + 4) \end{vmatrix} = 0 \Rightarrow \lambda = 5$

Now $D_x = \begin{vmatrix} 9 & 3 & 5 \\ 8 & 3 & -2 \\ 16 - \mu & 3 & -9 \end{vmatrix} = 0 \Rightarrow \mu = 9$

So centre of circle (5, 9)

So $r = \frac{|20 - 27|}{\sqrt{4^2 + 3^2}} = \frac{7}{5}$

68. Let the system of equations

$$x + 5y - z = 1$$

$$4x + 3y - 3z = 7$$

$$24x + y + \lambda z = \mu$$

$\lambda, \mu \in \mathbb{R}$, have infinitely many solutions. Then the number of the solutions of this system, if x, y, z are integers and satisfy $7 \leq x + y + z \leq 77$, is :

माना समीकरण निकाय

$$x + 5y - z = 1$$

$$4x + 3y - 3z = 7$$

$$24x + y + \lambda z = \mu$$

$\lambda, \mu \in \mathbb{R}$, के अनन्त हल हैं। तो इस निकाय के हलों, यदि x, y, z पूर्णांक हैं और $7 \leq x + y + z \leq 77$ को संतुष्ट करते हैं, की संख्या है :

(A) 5

(B) 3

(C) 6

(D) 4

Ans. B

Sol. $x + 5y - z = 1$... (1)

$$4x + 3y - 3z = 7$$
 ... (2)

$$24x + y + \lambda z = \mu$$
 ... (3)

put $z = k$ in equation (1) & (2)

$$x + 5y = 1 + k \quad \dots(4)$$

$$\text{and } 4x + 3y = 7 + 3k \quad \dots(5)$$

From (4) and (5)

$$x = \frac{32+12k}{17} \text{ and } y = \frac{k-3}{17}$$

$$\text{now, } x + y + z = \frac{32+12k}{17} + \frac{k-3}{17} + k$$

$$= \frac{29+30k}{17}$$

$$\therefore 7 \leq x + y + z \leq 77$$

$$\Rightarrow 7 \leq \frac{29+30k}{17} \leq 77$$

$$\Rightarrow 119 - 29 \leq 30k \leq 1309 - 29$$

$$\Rightarrow 3 \leq k \leq 42.6$$

for x, y, z to be integer also

$$k = 3, 20, 37$$

69. Let $A = \begin{bmatrix} 2 & 2+p & 2+p+q \\ 4 & 6+2p & 8+3p+2q \\ 6 & 12+3p & 20+6p+3q \end{bmatrix}$. If $\det(\text{adj}(\text{adj}(3A))) = 2^m \cdot 3^n$, $m, n \in \mathbb{N}$, then $m + n$ is equal to :

माना $A = \begin{bmatrix} 2 & 2+p & 2+p+q \\ 4 & 6+2p & 8+3p+2q \\ 6 & 12+3p & 20+6p+3q \end{bmatrix}$ है। यदि $\det(\text{adj}(\text{adj}(3A))) = 2^m \cdot 3^n$, $m, n \in \mathbb{N}$ है, तो $m + n$ बराबर है :

(A) 24

(B) 22

(C) 20

(D) 26

Ans. A

Sol. $|A| = \begin{vmatrix} 2 & 2+p & 2+p+q \\ 0 & 2 & 4+p \\ 0 & 6 & 14+3p \end{vmatrix} \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{matrix}$

$$|A| = 2(28 + 6p - 24 - 6p) = 8$$

Now,

$$|\text{adj}(\text{adj}(3A))| = |3A|^{(3-1)^2} = |3A|^4 = (3^3)^4 = 3^{12} \cdot 2^{12}$$

$$m + n = 12 + 12 = 24$$

70. If $\sum_{r=1}^n T_r = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{64}$, then $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{T_r} \right)$ is equal to :

यदि $\sum_{r=1}^n T_r = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{64}$ है, तो $\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{1}{T_r} \right)$ बराबर है :

(A) $\frac{1}{3}$ (B) 1 (C) $\frac{2}{3}$ (D) 0

Ans. C

Sol. $T_n = S_n - S_{n-1}$

$$= \frac{(2n-1)(2n+1)(2n+3)(2n+5) - (2n-3)(2n-1)(2n+1)(2n+3)}{64}$$

$$T_n = \frac{(2n-1)(2n+1)(2n+3)}{8}$$

$$\frac{1}{T_n} = \frac{8}{(2n-1)(2n+1)(2n+3)}$$

$$\frac{1}{T_n} = 2 \left(\frac{1}{T_{n-1}} - \frac{1}{T_n} \right)$$

$$S_n = 2 \left(\frac{1}{1 \times 3} - \frac{1}{(2n-1)(2n+3)} \right)$$

$$\lim_{n \rightarrow \infty} S_n = \frac{2}{3}$$

71. Let the system of equations

$$x + 5y - z = 1$$

$$4x + 3y - 3z = 7$$

$$24x + y + \lambda z = \mu$$

$\lambda, \mu \in \mathbb{R}$, have infinitely many solutions. Then the number of the solutions of this system, if x, y, z are integers and satisfy $7 \leq x + y + z \leq 77$, is :

माना समीकरण निकाय

$$x + 5y - z = 1$$

$$4x + 3y - 3z = 7$$

$$24x + y + \lambda z = \mu$$

$\lambda, \mu \in \mathbb{R}$, के अनन्त हल हैं। तो इस निकाय के हलों, यदि x, y, z पूर्णांक हैं और $7 \leq x + y + z \leq 77$ को संतुष्ट करते हैं, की संख्या है :

(A) 5

(B) 3

(C) 6

(D) 4

Ans. B

Sol. $x + 5y - z = 1$... (1)

$$4x + 3y - 3z = 7$$
 ... (2)

$$24x + y + \lambda z = \mu$$
 ... (3)

put $z = k$ in equation (1) & (2)

$$x + 5y = 1 + k$$
 ... (4)

$$\text{and } 4x + 3y = 7 + 3k$$
 ... (5)

From (4) and (5)

$$x = \frac{32 + 12k}{17} \text{ and } y = \frac{k - 3}{17}$$

$$\text{now, } x + y + z = \frac{32 + 12k}{17} + \frac{k - 3}{17} + k$$

$$= \frac{29 + 30k}{17}$$

$$\therefore 7 \leq x + y + z \leq 77$$

$$\Rightarrow 7 \leq \frac{29 + 30k}{17} \leq 77$$

$$\Rightarrow 119 - 29 \leq 30k \leq 1309 - 29$$

$$\Rightarrow 3 \leq k \leq 42.6$$

for x, y, z to be integer also

$$k = 3, 20, 37$$

72. If the system of linear equations

$$3x + y + \beta z = 3$$

$$2x + \alpha y - z = -3$$

$$x + 2y + z = 4$$

has infinitely many solutions, then the value of $22\beta - 9\alpha$ is :

यदि रैखिक समीकरण निकाय

$$3x + y + \beta z = 3$$

$$2x + \alpha y - z = -3$$

$$x + 2y + z = 4$$

के अनंत हल हैं, तो $22\beta - 9\alpha$ का मान है :

(A) 31

(B) 49

(C) 43

(D) 37

Ans. A

Sol. $\Delta = \begin{vmatrix} 3 & 1 & \beta \\ 2 & \alpha & -1 \\ 1 & 2 & 1 \end{vmatrix} = 0$

$$3\alpha + 4\beta - \alpha\beta + 3 = 0$$

$$\Delta_3 = \begin{vmatrix} 3 & 1 & 3 \\ 2 & \alpha & -3 \\ 1 & 2 & 4 \end{vmatrix} = 0$$

$$9\alpha + 19 = 0$$

$$\alpha = -\frac{19}{9}, \beta = \frac{6}{11}$$

$$\Rightarrow 22\beta - 9\alpha = 31$$

73. Let the system of equations :

$$2x + 3y + 5z = 9,$$

$$7x + 3y - 2z = 8,$$

$$12x + 3y - (4 + \lambda)z = 16 - \mu,$$

have infinitely many solutions. Then the radius of the circle centred at (λ, μ) and touching the line $4x = 3y$ is :

माना समीकरण निकाय

$$2x + 3y + 5z = 9,$$

$$7x + 3y - 2z = 8,$$

$$12x + 3y - (4 + \lambda)z = 16 - \mu,$$

के अनन्त हल हैं। तो उस वृत्त, जिसका केन्द्र (λ, μ) है और जो रेखा $4x = 3y$ को स्पर्श करता है, की त्रिज्या है :

- (A) 7 (B) $\frac{17}{5}$ (C) $\frac{21}{5}$ (D) $\frac{7}{5}$

Ans. D

Sol.
$$D = \begin{vmatrix} 2 & 3 & 5 \\ 4 & 3 & -2 \\ 12 & 3 & -(\lambda + 4) \end{vmatrix} = 0 \Rightarrow \lambda = 5$$

Now
$$D_x = \begin{vmatrix} 9 & 3 & 5 \\ 8 & 3 & -2 \\ 16 - \mu & 3 & -9 \end{vmatrix} = 0 \Rightarrow \mu = 9$$

So centre of circle $(5, 9)$

So
$$r = \frac{|20 - 27|}{\sqrt{4^2 + 3^2}} = \frac{7}{5}$$

INTEGER TYPE QUESTIONS

74. Let $A = I_2 - 2MM^T$, where M is a real matrix of order 2×1 such that the relation $M^T M = I_1$ holds. If λ is a real number such that the relation $AX = \lambda X$ holds for some non-zero real matrix X of order 2×1 , then the sum of squares of all possible values of λ is equal to _____.

माना 2×1 के एक वास्तविक आव्यूह M के लिए $M^T M = I_1$ है तथा $A = I_2 - 2MM^T$ है। λ एक वास्तविक संख्या है तथा कोटि 2×1 के किसी शून्येतर वास्तविक आव्यूह X के लिए $AX = \lambda X$ है, तो λ के सभी संभव मानों के वर्गों का योग है _____

Ans. 2

Sol.
$$\begin{aligned} A &= I_2 - 2MM^T \\ A^2 &= (I_2 - 2MM^T)(I_2 - 2MM^T) \\ &= I_2 - 2MM^T - 2MM^T + 4MM^TMM^T \\ &= I_2 - 4MM^T + 4MM^T \\ &= I_2 \\ AX &= \lambda X \\ A^2X &= \lambda AX \\ X &= \lambda(\lambda X) \end{aligned}$$

$$X = \lambda^2 X$$

$$X(\lambda^2 - 1) = 0$$

$$\lambda^2 = 1 \Rightarrow \lambda^2 = 1$$

$$\lambda = \pm 1$$

Sum of square of all possible values = 2

75. Consider the matrices : $A = \begin{bmatrix} 2 & -5 \\ 3 & m \end{bmatrix}$, $B = \begin{bmatrix} 20 \\ m \end{bmatrix}$ and $X = \begin{bmatrix} x \\ y \end{bmatrix}$. Let the set of all m, for which the system of equations $AX = B$ has a negative solution (i.e., $x < 0$ and $y < 0$), be the interval (a, b). Then $8 \int_a^b |A| dm$ is _____.

आव्यूहों $A = \begin{bmatrix} 2 & -5 \\ 3 & m \end{bmatrix}$, $B = \begin{bmatrix} 20 \\ m \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ का विचार कीजिए। माना m के सभी मान, जिनके लिए समीकरण निकाय $AX =$

B का ऋणात्मक हल (अर्थात्, $x < 0$ और $y < 0$) का समुच्चय, अंतराल (a, b) है। तो $8 \int_a^b |A| dm$ बराबर है _____.

Ans. 450

Sol. $A = \begin{pmatrix} 2 & -5 \\ 3 & m \end{pmatrix}$, $B = \begin{pmatrix} 20 \\ m \end{pmatrix}$

$$X = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$2x - 5y = 20 \quad \text{_____ (1)}$$

$$3x + my = m \quad \text{_____ (2)}$$

$$\Rightarrow y = \frac{2m - 60}{2m + 15}$$

$$y < 0 \Rightarrow m \in \left(\frac{-15}{2}, 30 \right)$$

$$x = \frac{25m}{2m + 15}$$

$$x < 0 \Rightarrow m \in \left(\frac{-15}{2}, 0 \right)$$

$$|A| = 2m + 15$$

$$\text{Now, } 8 \int_{-15}^0 (2m + 15) dm = 8 \left\{ m^2 + 15m \right\}_{-15}^0$$

$$\Rightarrow 8 \left\{ - \left(\frac{225}{4} - \frac{225}{2} \right) \right\}$$

$$= 8 \times \frac{225}{4} = 450$$

76. If the sum of the first 10 terms of the series $\frac{4 \cdot 1}{1 + 4 \cdot 1^4} + \frac{4 \cdot 2}{1 + 4 \cdot 2^4} + \frac{4 \cdot 3}{1 + 4 \cdot 3^4} + \dots$ is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $m + n$ is equal to _____.

यदि श्रेणी $\frac{4 \cdot 1}{1 + 4 \cdot 1^4} + \frac{4 \cdot 2}{1 + 4 \cdot 2^4} + \frac{4 \cdot 3}{1 + 4 \cdot 3^4} + \dots$ के प्रथम 10 पदों का योग $\frac{m}{n}$, $\gcd(m, n) = 1$, है, तो $m + n$ बराबर है _____.

Ans. 441

Sol. $\frac{4 \cdot 1}{1 + 4 \cdot 1^4} + \frac{4 \cdot 2}{1 + 4 \cdot 2^4} + \frac{4 \cdot 3}{1 + 4 \cdot 3^4} + \dots$

$$T_r = \frac{4r}{1 + 4r^4} = \frac{4r}{4r^4 + 4r^2 + 1 - 4r^2}$$

$$= \frac{4r}{(2r^2 + 1)^2 - (2r)^2}$$

$$T_r = \frac{4r}{(2r^2 - 2r + 1)(2r^2 + 2r + 1)}$$

$$T_r = \frac{(2r^2 + 2r + 1) - (2r^2 - 2r + 1)}{(2r^2 - 2r + 1)(2r^2 + 2r + 1)}$$

$$T_r = \left(\frac{1}{r^2 + (r-1)^2} - \frac{1}{r^2 + (r+1)^2} \right)$$

$$\sum_{r=1}^{10} T_r = \left(\frac{1}{0^2 + 1^2} - \frac{1}{1^2 + 2^2} + \frac{1}{1^2 + 2^2} - \frac{1}{2^2 + 3^2} + \dots - \frac{1}{9^2 + 10^2} - \frac{1}{10^2 + 11^2} \right)$$

$$= 1 - \frac{1}{221}$$

$$= \frac{220}{221}$$

$$\therefore m + n = 220 + 221$$

$$= 441$$

77. Let A be a 3×3 matrix such that $X^T A X = O$ for all nonzero 3×1 matrices $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. If

$$A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -5 \end{bmatrix}, A \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -8 \end{bmatrix}, \text{ and } \det(\text{adj}(2(A + I))) = 2^\alpha 3^\beta 5^\gamma, \alpha, \beta, \gamma \in \mathbb{N}, \text{ then } \alpha^2 + \beta^2 + \gamma^2 \text{ is } \underline{\hspace{2cm}}.$$

माना कि A एक 3×3 आव्यूह इस प्रकार है कि सभी शून्येतर 3×1 आव्यूहों $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ के लिए $X^T A X = O$ है। यदि

$$A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -5 \end{bmatrix}, A \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -8 \end{bmatrix}, \text{ और } \det(\text{adj}(2(A + I))) = 2^\alpha 3^\beta 5^\gamma, \alpha, \beta, \gamma \in \mathbb{N} \text{ है, तो } \alpha^2 + \beta^2 + \gamma^2 \text{ है } \underline{\hspace{2cm}}.$$

Ans. 44

Sol. $X^T A X = O$

$$(x \ y \ z) \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = O$$

$$(x \ y \ z) \begin{pmatrix} a_1 x + a_2 y + a_3 z \\ b_1 x + b_2 y + b_3 z \\ c_1 x + c_2 y + c_3 z \end{pmatrix} = O$$

$$x(a_1 x + a_2 y + a_3 z) + y(b_1 x + b_2 y + b_3 z) + z(c_1 x + c_2 y + c_3 z) = O$$

$$a_1 = 0, b_2 = 0, c_3 = 0$$

$$a_2 + b_1 = 0, a_3 + c_1 = 0, b_3 = c_2 = 0$$

$A = \text{skew symmetric matrix}$

$$A = \begin{pmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{pmatrix}; A = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ -5 \end{pmatrix}$$

$$\Rightarrow A = \begin{pmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ -5 \end{pmatrix}$$

$$x + y = 1$$

$$-x + z = 4$$

$$y + z = 5$$

$$\begin{pmatrix} 0 & x & y \\ -x & 0 & z \\ -y & -z & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ -8 \end{pmatrix}$$

$$2x + y = 0 \quad x = -1$$

$$-x + z = 4 \quad y = 2$$

$$-y - 2z = -8 \quad z = 3$$

$$A = \begin{pmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{pmatrix}$$

$$2(A + I) = \begin{pmatrix} 2 & -2 & 4 \\ 2 & 2 & 6 \\ -2 & -6 & 2 \end{pmatrix}$$

$$2(A + I) = 120 \Rightarrow \det[\text{adj}(2(A + I))]$$

$$= 120^2 = 2^6 \cdot 3^2 \cdot 5^2$$

$$\alpha = 6, \beta = 2, \gamma = 2$$

78. Let $\alpha = 1^2 + 4^2 + 8^2 + 13^2 + 19^2 + 26^2 + \dots$ upto 10 terms and $\beta = \sum_{n=1}^{10} n^4$. If $4\alpha - \beta = 55k + 40$, then k is equal to _____.

माना $\alpha = 1^2 + 4^2 + 8^2 + 13^2 + 19^2 + 26^2 + \dots$ 10 पदों तक तथा $\beta = \sum_{n=1}^{10} n^4$ हैं। यदि $4\alpha - \beta = 55k + 40$ है, तो k बराबर है _____।

Ans. 353

Sol. $\alpha = 1^2 + 4^2 + 8^2 \dots$

$$t_n = an^2 + bn + c$$

$$1 = a + b + c$$

$$4 = 4a + 2b + c$$

$$8 = 9a + 3b + c$$

On solving we get, $a = \frac{1}{2}, b = \frac{3}{2}, c = -1$

$$\alpha = \sum_{n=1}^{10} \left(\frac{n^2}{2} + \frac{3n}{2} - 1 \right)^2$$

$$4\alpha = \sum_{n=1}^{10} (n^2 + 3n - 2)^2, \beta = \sum_{n=1}^{10} n^4$$

$$4\alpha - \beta = \sum_{n=1}^{10} (6n^3 + 5n^2 - 12n + 4) = 55(353) + 40$$

79. Let S be the set of values of λ , for which the system of equations

$$6\lambda x - 3y + 3z = 4\lambda^2,$$

$$2x + 6\lambda y + 4z = 1,$$

$$3x + 2y + 3\lambda z = \lambda$$

has no solution. Then $12 \sum_{\lambda \in S} |\lambda|$ is equal to _____.

माना λ के मानों, जिनके लिए समीकरण निकाय

$$6\lambda x - 3y + 3z = 4\lambda^2,$$

$$2x + 6\lambda y + 4z = 1,$$

$$3x + 2y + 3\lambda z = \lambda$$

का कोई हल नहीं है, का समुच्चय S है। तो $12 \sum_{\lambda \in S} |\lambda|$ बराबर है _____

Ans. 24

Sol. For no solution

$$\begin{vmatrix} 6\lambda & -3 & 3 \\ 2 & 6\lambda & 4 \\ 3 & 2 & 3\lambda \end{vmatrix} = 0$$

$$\Rightarrow 9\lambda^3 - 7\lambda - 2 = 0$$

$$\Rightarrow (\lambda - 1)(3\lambda + 1)(3\lambda + 2) = 0$$

$$\Rightarrow 12 \sum_{\lambda \in S} |\lambda| = 12 \times \left(1 + \frac{1}{3} + \frac{2}{3}\right) = 24$$

80. Let $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$. If $\sum_{k=1}^n D_k = 96$, then n is equal to _____.

माना $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$ है। यदि $\sum_{k=1}^n D_k = 96$ तो n बराबर है _____

Ans. 6

Sol. $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$

$$\sum_{k=1}^n D_k = 96 \Rightarrow$$

$$\begin{vmatrix} \sum_{k=1}^n 1 & \sum_{k=1}^n 2k & \sum_{k=1}^n (2k-1) \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix} = 96$$

$$\Rightarrow \begin{vmatrix} n & n^2+n & n^2 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix} = 96$$

$$R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1$$

$$\begin{vmatrix} n & n^2+n & n^2 \\ 0 & 2 & 0 \\ 0 & 0 & n+2 \end{vmatrix} = 96$$

$$\Rightarrow n(2n+4) = 96 \Rightarrow n(n+2) = 48 \Rightarrow n = 6$$